

19.

Given $U_4 = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2}$

$$\frac{dU_4}{d\beta} = \frac{-1}{3} \frac{d}{d\beta} \left(\frac{\langle m^4 \rangle}{\langle m^2 \rangle^2} \right) = -\frac{1}{3} \left[\frac{d}{d\beta} (\langle m^4 \rangle) \frac{1}{\langle m^2 \rangle^2} - \frac{2\langle m^4 \rangle}{\langle m^2 \rangle^3} \frac{d}{d\beta} (\langle m^2 \rangle) \right]$$

$$\frac{d\langle m^4 \rangle}{d\beta} = \frac{d}{d\beta} \left(\frac{\sum m^4 e^{-\beta H}}{\sum e^{-\beta H}} \right) = \frac{\sum -m^4 H e^{-\beta H}}{\sum e^{-\beta H}} - \frac{\sum m^4 e^{-\beta H} \sum -H e^{-\beta H}}{(\sum e^{-\beta H})^2}$$

$= -\langle m^4 H \rangle + \langle m^4 \rangle \langle H \rangle$

$$\Rightarrow \frac{d\langle m^4 \rangle}{d\beta} = -\langle m^4 H \rangle + \langle m^4 \rangle \langle H \rangle$$

$$\frac{2\langle m^4 \rangle}{\langle m^2 \rangle^3} \frac{d\langle m^2 \rangle}{d\beta} = \frac{2\langle m^4 \rangle}{\langle m^2 \rangle^2} \frac{d \ln(\langle m^2 \rangle)}{d\beta}$$

$$\Rightarrow \frac{2\langle m^4 \rangle}{\langle m^2 \rangle^3} \frac{d\langle m^2 \rangle}{d\beta} = -\frac{2\langle m^4 \rangle \langle m^2 H \rangle}{\langle m^2 \rangle^3} + \frac{2\langle m^4 \rangle \langle H \rangle \langle m^2 \rangle}{\langle m^2 \rangle^3}$$

$$\Rightarrow \frac{dU_4}{d\beta} = -\frac{1}{3} \left[\frac{-\langle m^4 H \rangle + \langle m^4 \rangle \langle H \rangle}{\langle m^2 \rangle^2} - \frac{2\langle H \rangle \langle m^4 \rangle}{\langle m^2 \rangle^2} + \frac{2\langle m^4 \rangle \langle m^2 H \rangle}{\langle m^2 \rangle^3} \right]$$

$$\frac{dU_4}{d\beta} = -\frac{1}{3} \left[\frac{1}{\langle m^2 \rangle^2} \left[\frac{2\langle m^4 \rangle \langle m^2 H \rangle}{\langle m^2 \rangle} - \langle m^4 H \rangle - \langle H \rangle \langle m^4 \rangle \right] \right]$$

$$\Rightarrow \frac{dU_4}{d\beta} = -\frac{1}{3} \left[\frac{\langle m^4 \rangle}{\langle m^2 \rangle^2} \left[\frac{2\langle m^2 H \rangle}{\langle m^2 \rangle} - \frac{\langle m^4 H \rangle}{\langle m^4 \rangle} - \langle H \rangle \right] \right]$$

$$\frac{dU_4}{d\beta} = \frac{1}{3} \frac{\langle m^4 \rangle}{\langle m^2 \rangle^2} \left(\langle H \rangle - 2 \frac{\langle m^2 H \rangle}{\langle m^2 \rangle} + \frac{\langle m^4 H \rangle}{\langle m^4 \rangle} \right)$$

$$\frac{1}{3} \frac{\langle m^2 \rangle}{\langle m^2 \rangle^2} = 1 - u_1$$

$$\Rightarrow \frac{d u_1}{d \beta} = (1 - u_1) \left(\langle H \rangle - 2 \frac{\langle m^2 H \rangle}{\langle m^2 \rangle} + \frac{\langle m^4 H \rangle}{\langle m^2 \rangle} \right)$$

For a Hamiltonian $H = eV$ where e is the energy densities and V the volume" ($V = L^2$)

$$\langle H \rangle = V \langle e \rangle$$

$$\langle H m^2 \rangle = V \langle A e \rangle$$

$$\Rightarrow \frac{d u_1}{d \beta} = V(1 - u_1) \left(\langle e \rangle - 2 \frac{\langle m^2 e \rangle}{\langle m^2 \rangle} + \frac{\langle m^4 e \rangle}{\langle m^2 \rangle} \right)$$

For $u_2 = 1 - \frac{\langle m^2 \rangle}{3 \langle m^4 \rangle}$ the result follows the same form.

$$\frac{d u_2}{d \beta} = V(1 - u_2) \left(\langle e \rangle - \frac{2 \langle m^2 e \rangle}{\langle m^2 \rangle} + \frac{\langle m^4 e \rangle}{\langle m^2 \rangle} \right)$$

b)

$$\frac{d \ln(\langle m^2 \rangle)}{d \beta} = \frac{d \langle m^2 \rangle}{d \beta} \frac{1}{\langle m^2 \rangle} = \frac{1}{\langle m^2 \rangle} \left(\langle m^2 \rangle \langle H \rangle - \langle m^2 H \rangle \right)$$

$$\frac{d \ln(\langle m^2 \rangle)}{d \beta} = \langle H \rangle - \frac{\langle m^2 H \rangle}{\langle m^2 \rangle} = V \left(\langle e \rangle - \frac{\langle m^2 e \rangle}{\langle m^2 \rangle} \right)$$

Similarly for $\langle m^4 \rangle$

$$\frac{d \ln \langle m^4 \rangle}{d \beta} = V \left(\langle e \rangle - \frac{\langle m^4 e \rangle}{\langle m^4 \rangle} \right)$$

HW10 - Computer Simulation II

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Problem 20

At a second-order transition, the different observables at $\beta = \beta_c$ scale as:

$$\chi(L) \sim L^{\gamma/\nu}, \quad C(L) \sim L^{\alpha/\nu}, \quad \left| \frac{dU_4}{d\beta} \right|_{\beta_c} \sim L^{1/\nu}, \quad (1)$$

with exact exponents $\gamma/\nu = 7/4$, $\alpha/\nu = 0$, and $1/\nu = 1$ for the Ising universal class. The Binder cumulants U_4 and U_2 converge to constants in the thermodynamic limit at $L \rightarrow \infty$.

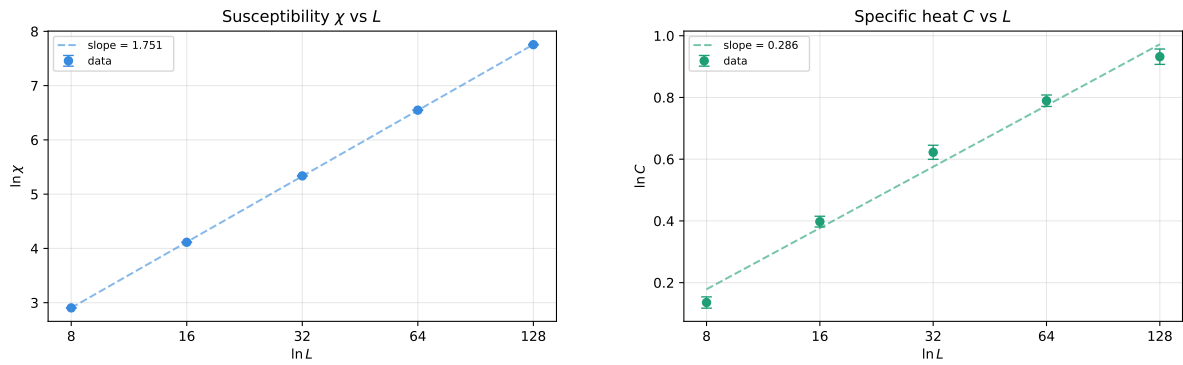


Figure 1: Left: susceptibility χ vs L in log-log scale. Right: specific heat C vs L .

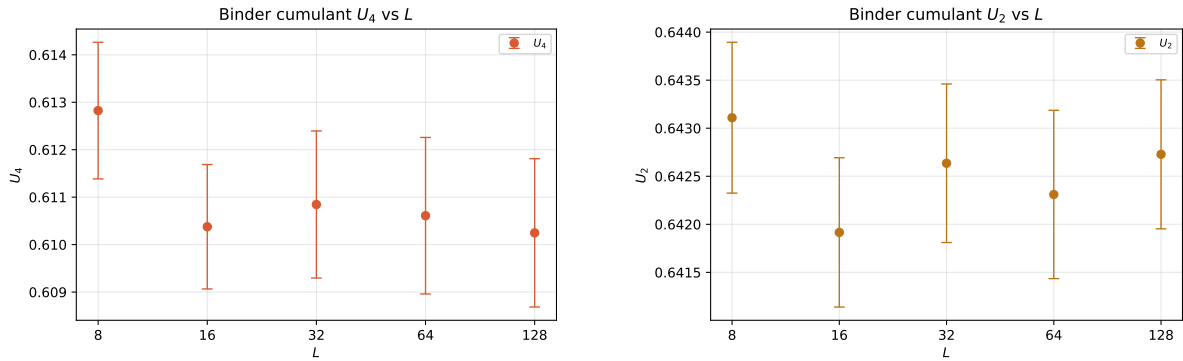


Figure 2: Left: Binder cumulants U_4 . Right: U_2 (right) vs L .

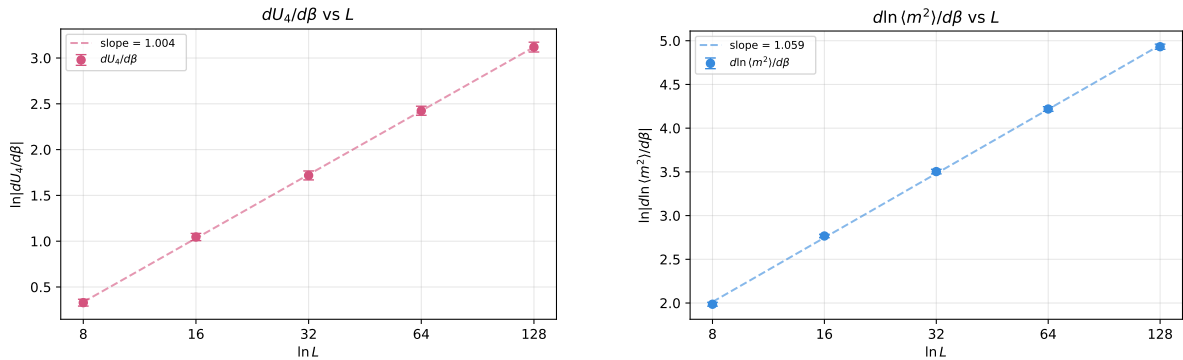


Figure 3: Left: Derivatives $dU_4/d\beta$. Right: $d\ln\langle m^2 \rangle/d\beta$ vs L in log-log scale.

Table 1: Measured observables at β_c for each lattice size L . Errors are standard deviation estimates.

L	C	χ	U_4	U_2	$dU_4/d\beta$	$d\ln\langle m^2 \rangle/d\beta$
8	1.13	18.2	0.612 ± 0.002	0.644 ± 0.002	1.40 ± 0.06	7.29 ± 0.41
16	1.46	61.8	0.613 ± 0.002	0.646 ± 0.002	2.66 ± 0.14	14.08 ± 1.02
32	1.87	204.1	0.608 ± 0.002	0.641 ± 0.002	5.91 ± 0.31	28.30 ± 2.80
64	2.16	694.2	0.609 ± 0.002	0.642 ± 0.002	11.74 ± 0.57	57.99 ± 4.12
128	2.65	2320	0.608 ± 0.002	0.641 ± 0.002	23.72 ± 1.18	118.1 ± 9.9

Table 2: Fitted log-log slopes versus exact FSS predictions.

Quantity	Fitted slope	Exact	Deviation
$\chi \sim L^{\gamma/\nu}$	1.749	1.750	0.06%
$C \sim L^{\alpha/\nu}$ (log divergence)	0.201	0 (log)	-
$dU_4/d\beta \sim L^{1/\nu}$	1.032	1.000	3.2%
$d\ln\langle m^2 \rangle/d\beta \sim L^{1/\nu}$	1.011	1.000	1.1%

The susceptibility exponent $\gamma/\nu = 1.749 \pm 0.003$ agrees with the exact value $7/4$. The specific heat grows slowly with L , consistent with the known logarithmic divergence ($\alpha = 0$) rather than a power law. Both derivative quantities scale as $L^{1/\nu}$ with $1/\nu \approx 1.02$, confirming $\nu = 1$ and validating the results from Problem 19.

The Binder cumulant U_4 hovers around 0.61 across all lattice sizes, consistent with the known universal value $U_4 \approx 0.611$. The small remaining L dependence is a finite size correction of order L^{-2} .